

ROZAS AP, Spain

WMO: 080080

Lat: 43.1114N

Lon: 7.4575W

Elev: 446

StdP: 96.08

Time Zone: 1.00 (EUC)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.7	-4.4	2.7	-1.2	-3.0	3.1	-0.3	10.4	8.1	8.8	8.1	0.8	120	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	30.1	20.1	27.8	19.5	25.7	18.5	21.3	28.0	20.1	26.2	19.1	24.6	2.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.5	25.0	17.7	13.4	23.0	17.0	12.8	21.4	63.9	27.6	59.4	26.3	56.0	24.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.2	6.0	DB	-5.9	35.0	2.3	2.5	-7.5	36.9	-8.9	38.4	-10.1	39.8	-11.8	41.6	(4)
(5)			WB	-6.2	24.0	2.4	1.2	-8.0	24.8	-9.4	25.5	-10.7	26.2	-12.5	27.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	6.0	6.9	7.9	10.5	12.3	16.1	18.7	18.4	17.0	13.3	9.3	7.3	(6)	
	DBStd	5.37	3.25	2.80	2.50	2.96	3.37	3.10	2.46	2.76	2.89	2.76	3.50	3.18	(7)	
	HDD10.0	492	128	91	74	28	16	1	0	0	0	5	54	94	(8)	
	HDD18.3	2415	382	319	322	235	190	80	25	33	59	157	271	342	(9)	
	CDD10.0	1227	4	5	10	43	86	184	270	261	211	108	34	11	(10)	
	CDD18.3	110	0	0	0	0	2	14	37	35	21	2	0	0	(11)	
	CDH23.3	1081	0	0	2	14	52	146	269	346	213	38	0	0	(12)	
	CDH26.7	327	0	0	0	0	14	47	78	117	63	7	0	0	(13)	
(14) Wind	WSAvg	2.1	2.2	2.3	2.8	2.7	2.2	2.1	2.0	1.9	1.8	1.7	1.8	2.1	(14)	
(15) Precipitation	PrecAvg	1046	133	93	82	103	79	42	31	35	66	135	137	119	(15)	
	PrecMax	1316	284	213	297	338	166	160	76	114	205	262	300	276	(16)	
	PrecMin	677	28	19	0	19	18	0	1	7	10	12	29	18	(17)	
	PrecStd	188	73	55	66	70	38	36	19	23	42	55	65	66	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	20.4	20.9	25.2	28.7	31.8	31.9	33.5	31.6	27.1	20.2	17.8	(19)	
		MCWB	12.1	11.6	11.4	14.8	17.8	21.2	21.2	21.1	19.7	16.2	15.4	12.0	(20)	
	2%	DB	13.3	15.7	17.2	22.4	23.9	27.4	28.6	29.5	28.3	23.8	17.6	16.1	(21)	
		MCWB	11.2	10.3	11.6	14.5	17.2	19.9	20.7	20.2	18.7	16.3	14.4	11.3	(22)	
	5%	DB	11.9	13.1	14.7	19.1	20.5	24.2	26.3	27.0	25.6	21.1	16.1	14.1	(23)	
		MCWB	9.9	10.1	10.2	13.1	15.1	18.5	19.7	19.4	18.2	15.6	13.6	11.1	(24)	
	10%	DB	10.7	11.7	12.4	16.2	18.1	21.8	24.0	24.5	23.0	18.7	14.5	12.5	(25)	
		MCWB	9.0	9.5	9.3	11.6	13.8	17.1	18.8	18.4	17.4	14.8	12.6	10.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	12.6	13.4	16.0	19.4	23.2	22.6	23.0	20.8	18.4	16.7	13.5	(27)	
		MCDB	14.0	16.6	19.2	23.2	26.1	30.7	29.1	29.3	28.0	22.7	18.9	15.6	(28)	
	2%	WB	11.6	11.7	11.9	14.7	17.5	20.4	21.1	20.9	19.6	17.0	15.3	12.2	(29)	
		MCDB	12.6	13.9	15.8	21.4	23.4	25.5	27.6	27.7	26.3	21.2	17.3	13.9	(30)	
	5%	WB	10.4	10.7	10.9	13.4	15.8	18.7	20.0	19.7	18.5	16.1	13.9	11.6	(31)	
		MCDB	11.5	13.1	14.0	18.5	19.8	23.8	25.6	25.8	24.5	20.0	15.7	13.5	(32)	
	10%	WB	9.2	9.6	9.8	12.1	14.1	17.4	18.9	18.6	17.5	15.1	12.7	10.7	(33)	
		MCDB	10.4	11.6	12.3	15.7	17.3	21.3	23.5	23.8	22.1	18.8	14.1	12.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.2	9.1	10.1	9.7	9.6	10.5	12.2	12.2	10.9	7.4	7.4	(35)	
		MCDBR	7.8	12.2	13.9	16.5	15.0	15.2	15.0	17.0	16.8	15.7	9.4	10.0	(36)	
	5% WB	MCWBR	5.8	7.8	8.4	9.3	8.6	8.2	7.8	8.3	7.7	9.1	6.4	6.9	(37)	
		MCWBR	5.8	10.6	12.2	15.1	14.6	13.5	13.8	15.7	15.0	13.6	8.1	8.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.325	0.358	0.369	0.376	0.381	0.368	0.368	0.360	0.348	0.324	0.309	(40)	
		taud	2.502	2.456	2.365	2.340	2.344	2.349	2.385	2.394	2.417	2.465	2.497	2.512	(41)	
	Ebn at Noon		824	874	881	897	897	890	898	887	868	834	804	796	(42)	
		Edh at Noon	71	87	108	119	122	122	117	112	101	86	71	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.45	2.32	3.45	4.56	5.48	6.11	6.18	5.62	4.41	2.79	1.67	1.33	(44)	
	RadStd		0.19	0.34	0.53	0.54	0.53	0.47	0.38	0.38	0.24	0.34	0.26	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page